Employee Workforce Segmentation

1. Executive Summary

This report segments 1,470 employees into 3 workforce groups using 22 HR features. Principal Component Analysis (PCA) reduced the features to 11 components (82.3% of variance); K-Means and Ward hierarchical clustering were then applied and compared. Attrition was held out of the clustering and used only to validate which segments are at risk.

Employees	Overall attrition	Segments	PCA components	High-risk cluster
1,470	14.2%	3	11 (82%)	C1

Key findings

- K-Means and Ward clustering broadly agree (Adjusted Rand Index 0.73); K-Means achieved the higher silhouette score.
- Segments differ significantly in attrition (chi-square p = 1.6e-13). Cluster 1 (748 employees) leaves at 19.5%, 1.37x the company average.
- An independent Retention Risk Index, built without the attrition label, reaches ROC-AUC 0.71. Employees in its High group leave at 24.7% versus 4.9% for the Low group.
- 66 top-performing employees left the company, a regretted-attrition cost worth targeting.

2. Data and Method

The analysis follows five steps:

- Feature engineering: OverTime encoded 0/1, BusinessTravel as an ordinal 0 to 2, MonthlyIncome log-transformed to reduce skew; all features standardised to z-scores.
- PCA on the standardised features, retaining enough components to explain at least 80% of variance.
- K-Means and Ward hierarchical clustering in the reduced PCA space, with k chosen by silhouette score (searched over k = 3 to 8).
- Segment profiling, with automatic names derived from each segment's most distinctive features.
- Risk analysis by cluster, department, performance group and an independent retention-risk index.

Features used (22): Age, DistanceFromHome, Education, EnvironmentSatisfaction, JobInvolvement, JobLevel, JobSatisfaction, MonthlyIncome, NumCompaniesWorked, PercentSalaryHike, PerformanceRating, RelationshipSatisfaction, StockOptionLevel, TotalWorkingYears, TrainingTimesLastYear, WorkLifeBalance, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager, OverTime_flag, BusinessTravel_ord.

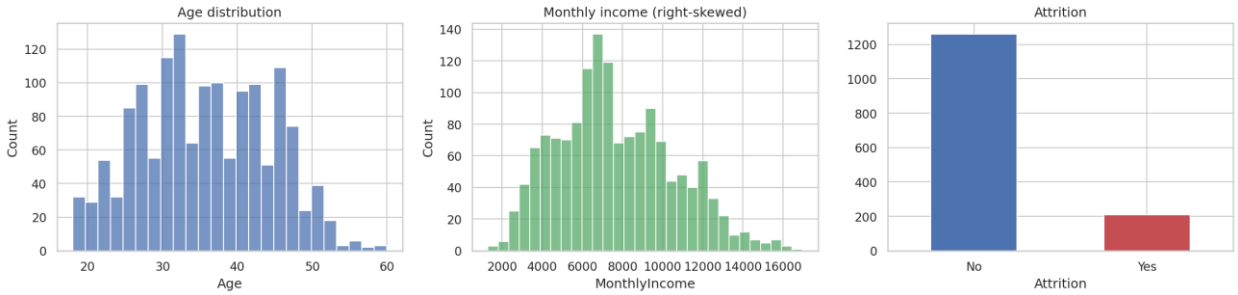


Figure 1. Age, income and attrition distributions

3. PCA Summary

11 principal components explain 82.3% of the total variance, compressing 22 original features by about half. The leading components are interpreted below.

Component	Explained variance	Cumulative
PC1	30.40%	30.40%
PC2	9.99%	40.39%
PC3	6.54%	46.93%
PC4	4.95%	51.88%
PC5	4.76%	56.65%
PC6	4.70%	61.35%
PC7	4.46%	65.81%
PC8	4.39%	70.19%

- **PC1:** main drivers are YearsAtCompany (+0.37), YearsInCurrentRole (+0.36), YearsWithCurrManager (+0.36), TotalWorkingYears (+0.35).
- **PC2:** main drivers are NumCompaniesWorked (+0.40), JobInvolvement (-0.39), JobSatisfaction (-0.36), PerformanceRating (-0.34).
- **PC3:** main drivers are PercentSalaryHike (+0.64), PerformanceRating (+0.59), JobInvolvement (-0.19), WorkLifeBalance (-0.19).

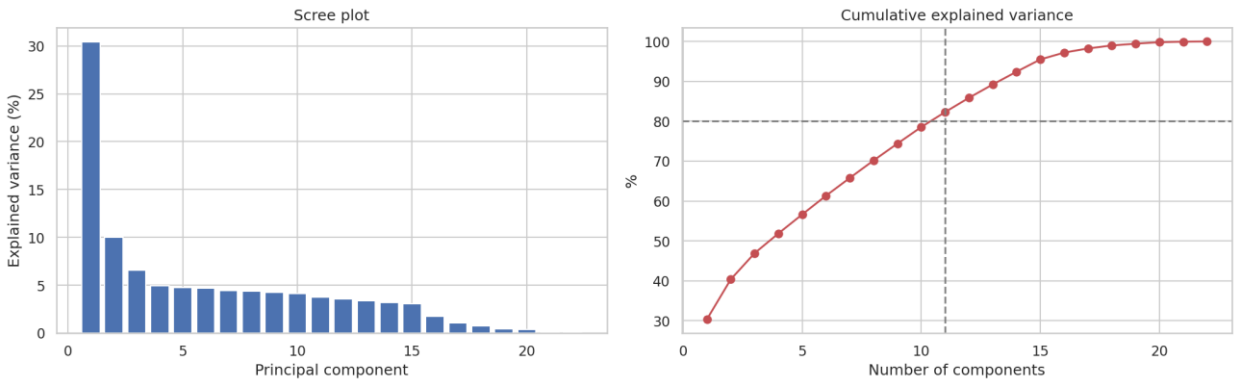


Figure 2. Scree plot and cumulative explained variance

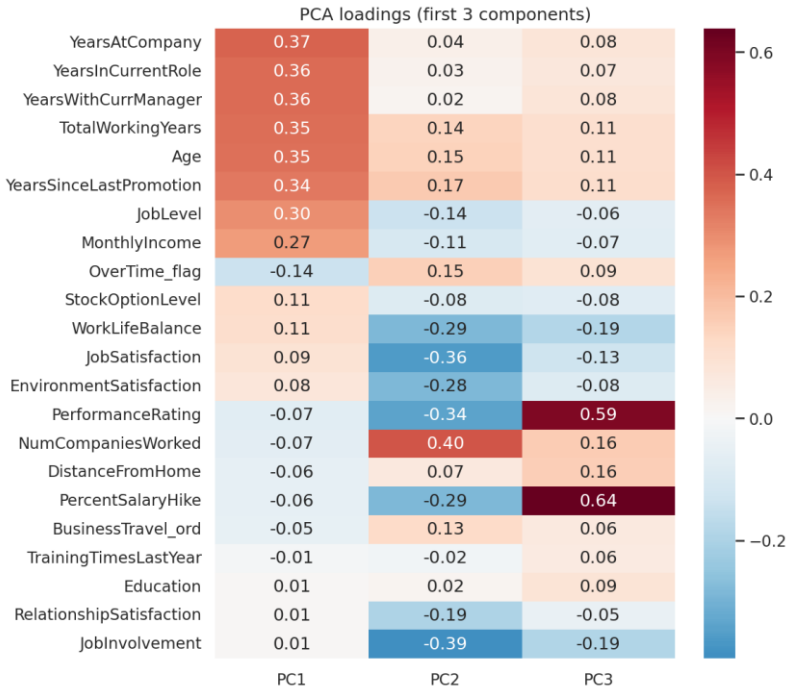


Figure 3. Loadings of the first three components

4. Clustering Results

4.1 Choosing the number of clusters

The silhouette score was highest at k = 3, so 3 segments were used.

k	Inertia	Silhouette	Davies-Bouldin
3	17092	0.169	1.876
4	15771	0.150	2.092
5	14894	0.136	2.102
6	14410	0.112	2.315
7	14010	0.114	2.197
8	13652	0.107	2.313

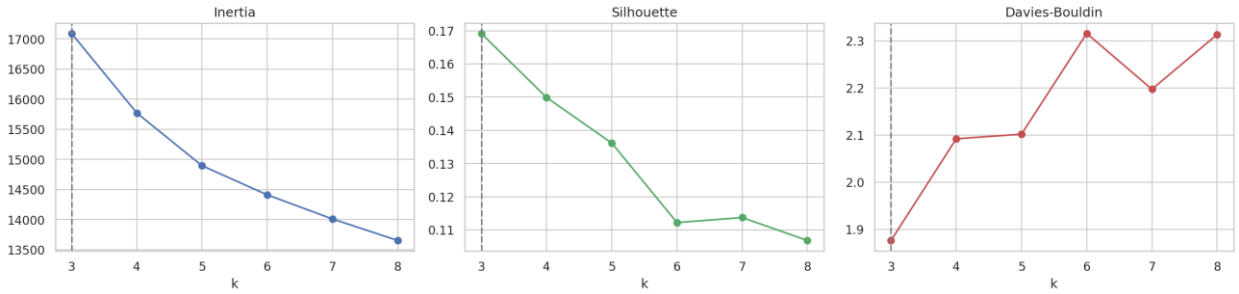


Figure 4. Inertia, silhouette and Davies-Bouldin by k

4.2 K-Means versus hierarchical clustering

Method	Silhouette	Davies-Bouldin
K-Means	0.169	1.876

Method	Silhouette	Davies-Bouldin
Hierarchical (Ward)	0.158	1.959

The two methods agree strongly (Adjusted Rand Index 0.73). K-Means is used as the primary segmentation; the dendrogram below confirms the hierarchical structure supports the same number of groups.

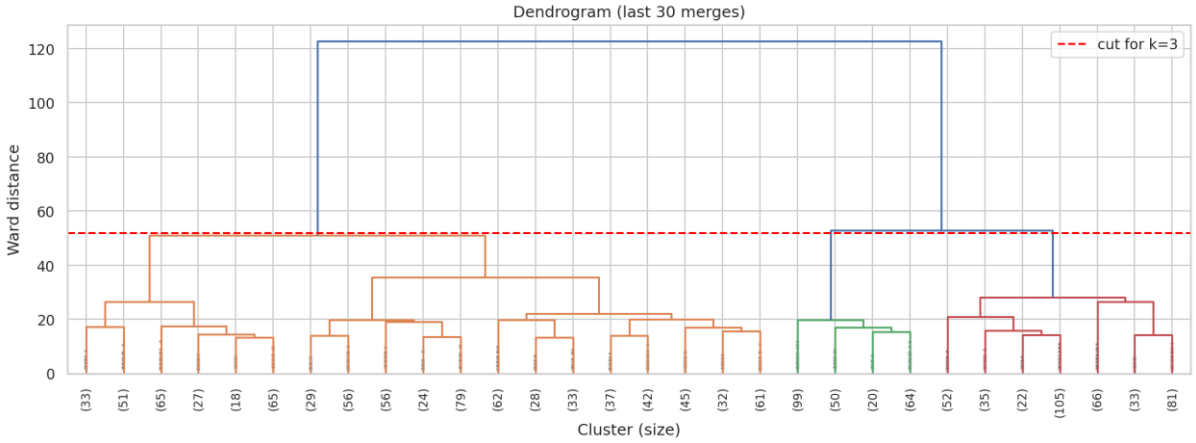


Figure 5. Ward dendrogram (last 30 merges) with the cut for the chosen k

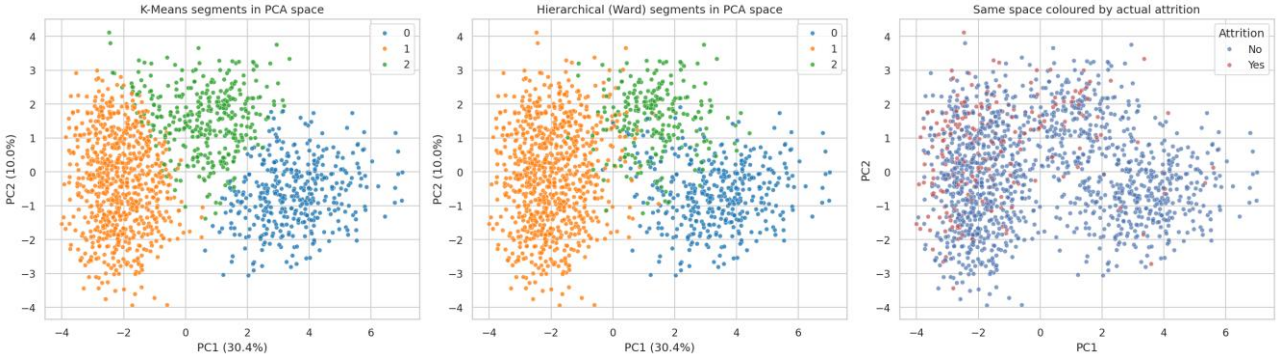


Figure 6. Segments in PCA space and actual attrition

5. Segment Profiles

Cluster	Segment profile	Head-count	% workforce	Avg income	Overtime	Attrition	Lift
0	C0: High YearsAtCompany, High YearsInCurrentRole	388	26.4%	10,920	16%	3.0%	0.20x
1	C1: Low YearsSinceLastPromotion, Low YearsAtCompany	748	50.9%	6,023	52%	20.0%	1.37x
2	C2: Low JobInvolvement, High NumCompaniesWorked	334	22.7%	7,511	30%	16.0%	1.10x

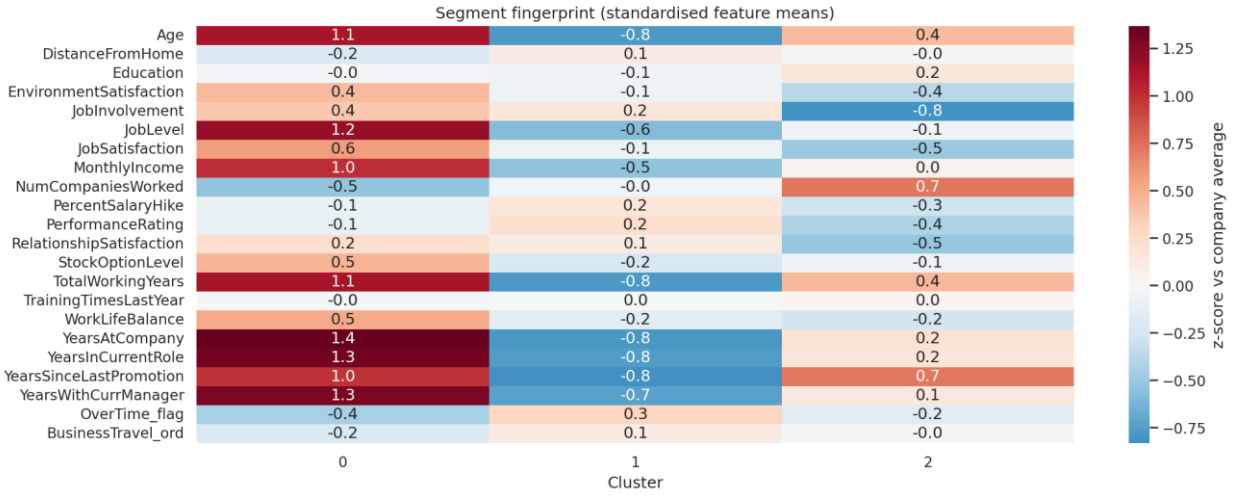


Figure 7. Segment fingerprint: standardised feature means by cluster

6. High-Risk Employee Clusters

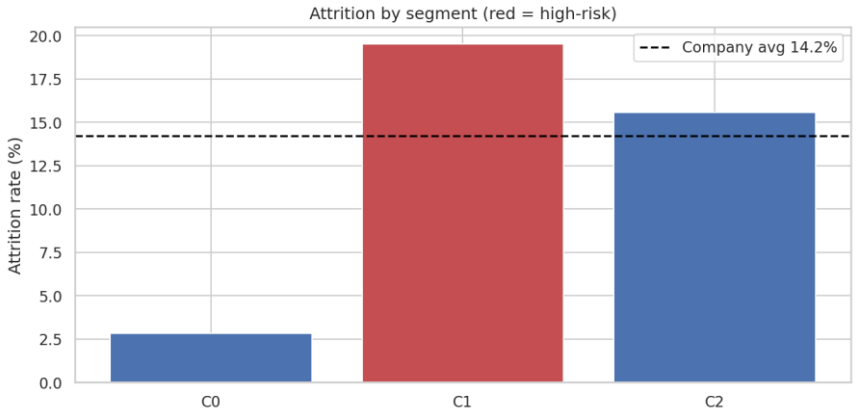


Figure 8. Attrition by segment (red = high-risk)

Cluster 1: 748 employees, attrition 19.5% (1.37x average). Distinguishing traits: low YearsSinceLastPromotion, low YearsAtCompany, low TotalWorkingYears, low Age.

7. Department-Level Workforce Segments

Department	Headcount	Attrition rate	Dominant segment	Share of dominant segment
Human Resources	114	14.9%	C1	46.5%
Research & Development	898	11.8%	C1	49.4%
Sales	458	18.8%	C1	54.8%

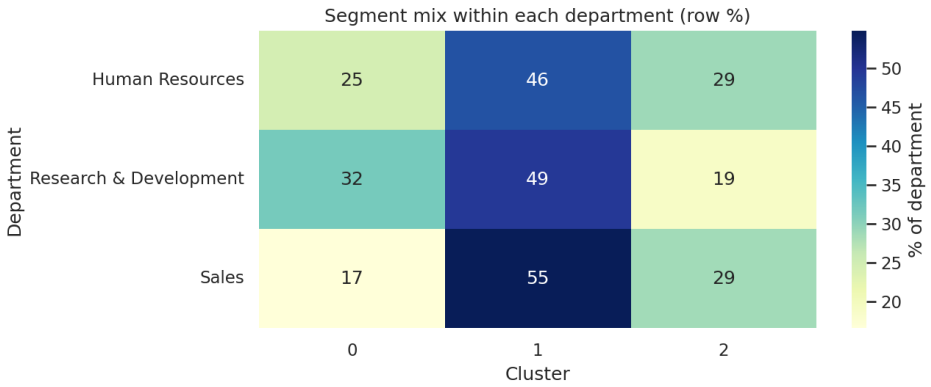


Figure 9. Segment mix within each department (row %)

8. Performance Groups

Employees are split into three equal groups using a composite of performance rating, salary hike and job involvement.

Performance group	Headcount	Attrition rate
Needs support	490	16.5%
Solid	490	12.7%
Top performer	490	13.5%

66 top performers left the company (regretted attrition).

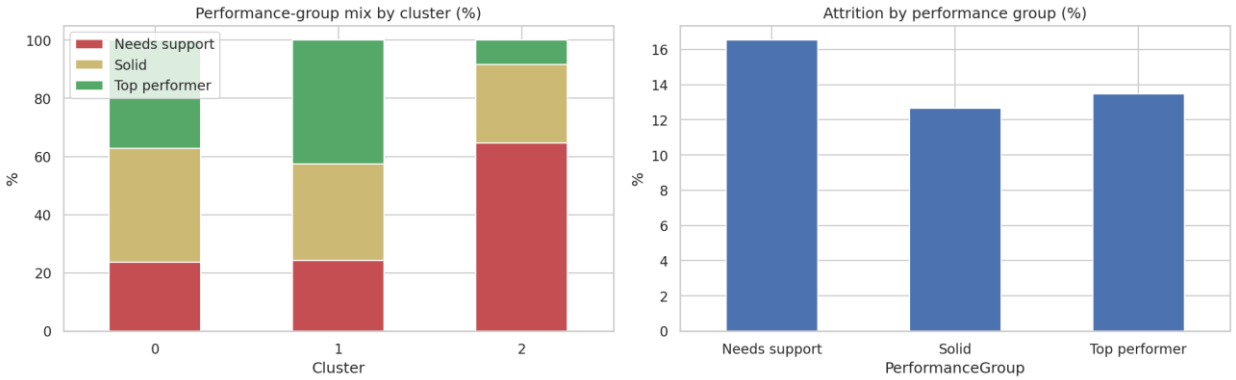


Figure 10. Performance-group mix by cluster and attrition by performance group

9. Retention-Risk Groups

The Retention Risk Index combines low satisfaction, poor work-life balance, few stock options, overtime, frequent travel, long time since promotion and pay below the job-level norm. It does not use the attrition label, so it can be validated against it (ROC-AUC 0.71).

Risk group	Headcount	Attrition rate	Avg risk index
Low	490	4.9%	-0.41
Medium	490	13.1%	-0.02
High	490	24.7%	0.42

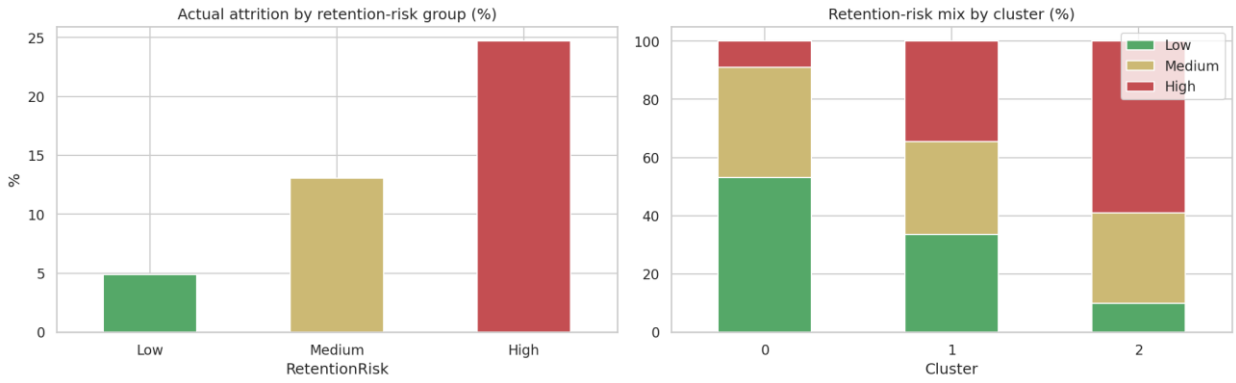


Figure 11. Attrition by risk group and risk mix by cluster

10. Workforce Analytics Dashboard

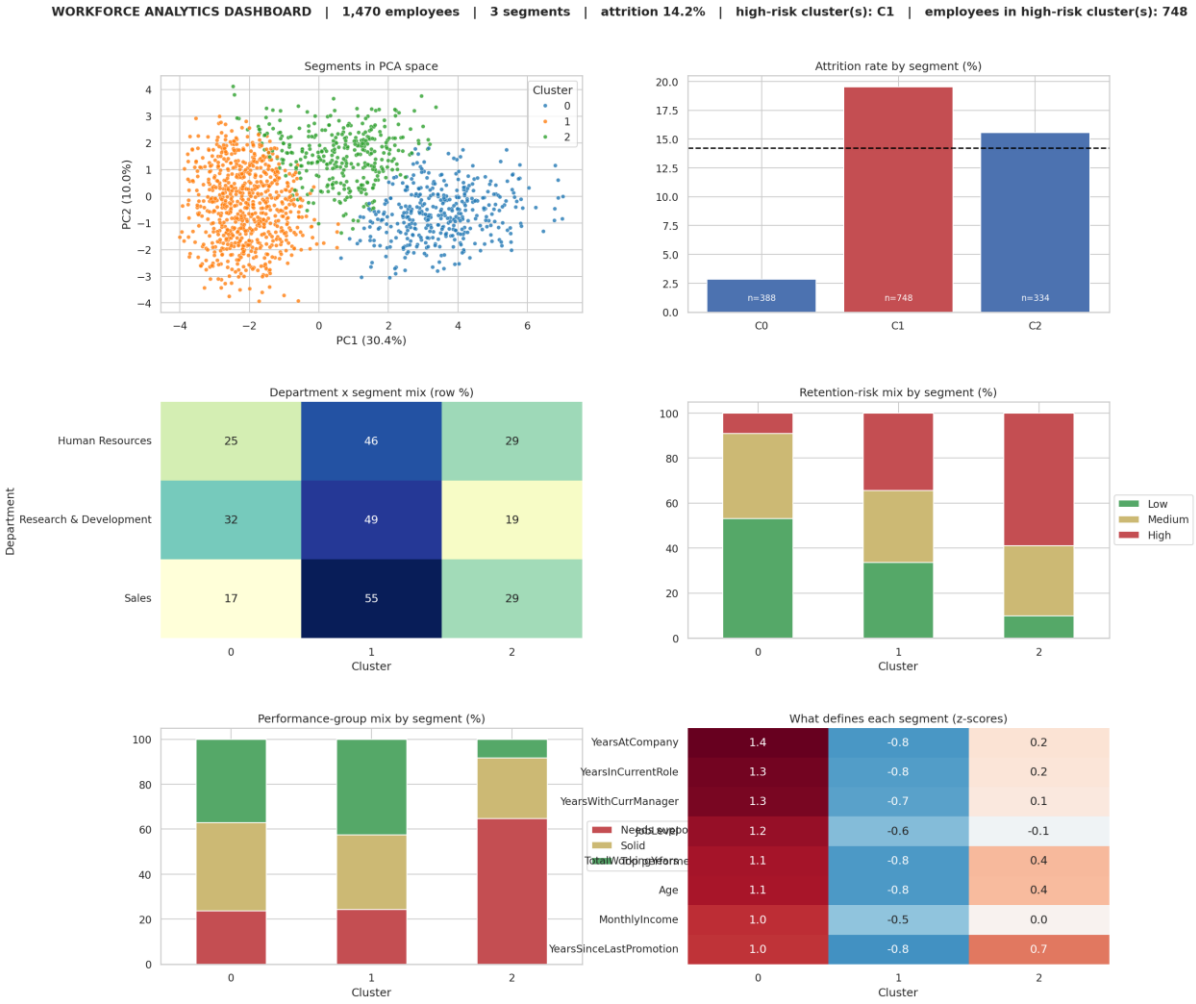


Figure 12. Consolidated workforce analytics dashboard

11. Recommendations

- Prioritise retention programmes for high-risk clusters and the High retention-risk group; target drivers such as workload, promotion stagnation and engagement.
- Create a watch-list of top performers sitting in High-risk groups and review their development and compensation.
- Use department-level segment mixes to tailor interventions rather than company-wide policies.

- Refresh the segmentation each quarter to track drift.

12. Limitations

- Clusters are descriptive, not causal; validate drivers with HR before acting.
- Segment names are auto-generated from the most distinctive features.
- Results here use synthetic data; re-run the notebook on the real dataset.
- Use results at group level and follow privacy and fairness policies; segment membership alone should not drive individual employment decisions.